

Course Description

Welcome to Epidemiology of Infectious Disease. We will begin by reviewing the course topics, measures of disease, frequency, and association that we will use in epidemiology, and the magnitude of impact of infectious disease on the world, with available measures. Targeting the students with less biology-background, individual pathogens are introduced and discussed at length based on their categories early on in the course, so that students can correlate the characteristics of a particular pathogen to the specific disease it is causing, which are discussed in the subsequent weeks. Overall, the diseases are organized in this course by modes of transmission – forms of person to person transmission in relation to the way infectious diseases spread in the environment. Targeted lectures on immunology and on antibiotics complement the overall objective of the course.

Learning Objectives

- Recognize the spectrum of infectious diseases according to organ system, and clinical syndrome.
- Become familiar with the separate biology (ecology, anatomy, physiology, and pathogenesis) of pathogenic bacteria, fungal, viral, and parasitic pathogens.
- Review the basic tools and study designs of epidemiology with respect to applications in the prevention of infectious disease.
- Obtain a basic knowledge of human immunobiology and the role it can play in the protection and exacerbation of infectious disease.
- Become familiar with the methods of development and distribution of anti-infection vaccines.
- Recognize the reasons for the crisis in antibiotic development and the potential solutions.
- Appreciate the different means by which human beings can become exposed to pathogenic agents.
- Understand how strategies of prevention and intervention are based on the biology of the pathogenic agents.

- Become able to recognize patterns of infectious disease syndrome occurrence in populations, use the literature to generate possible explanations for those patterns, and evaluate methods of short-term and long-term intervention.

Course Notes

Sessions and Days of the Week

The course is divided into fifteen weeks. Due dates for assignments and discussions are stated in day numbers. Day 1 is Wednesday, the first day of the beginning of each weekly session.

Class Day	Corresponding Day of the Week
Day 1	Wednesday
Day 2	Thursday
Day 3	Friday
Day 4	Saturday
Day 5	Sunday
Day 6	Monday
Day 7	Tuesday

Assignments are due no later than 11:55 p.m. in the Pacific Time zone on the day that is stated in the assignment page.

Class Session and/or contact hours weekly

Class contact time will be met by a combination of online asynchronous learning (e.g., recorded lectures, videos, expert interviews, moderated discussion forums) followed by online synchronous activities (e.g., required live sessions). In order to make as much time as possible for active engagement during the live sessions, you should prepare by doing all the reading, watching all the videos, and completing all required tasks in advance. The live sessions build directly on these experiences and cannot be duplicated outside of class.

This design is the “flipped” classroom model and is based on research that clearly demonstrates that learning is enhanced when working actively as opposed to listening passively (as in a lecture). This model uses a student-centered approach that emphasizes critical thinking, problem-solving and active learning.

Courses will have 10-15 live sessions. For more details on the days/times of these sessions, visit your CANVAS Course Overview Page.

Communication

Your instructors will primarily communicate with you via email, regular course announcements, and in live sessions. Weekly office hours will not be scheduled; however, individual or group meeting times and ad hoc office hours may be scheduled or requested via email. Contact information will be posted to the Instructor Announcements discussion board.

Technological Proficiency and Hardware/Software Required

Canvas is where content, grades, and communication will reside for this course.

- <https://uscpublichealth.instructure.com/>

Navigation Tips: Use the Help button in the global Canvas navigation (left-side navigation menu) for Canvas Support.

For issues with Canvas, contact the technical support hotline:

- 1 (866) 452-2178
- <https://community.canvaslms.com/t5/Guides/ct-p/guides>

USC Technology Support Links

[Zoom information for students](#)

[Blackboard help for students](#)

[Software available to USC Campus](#)

MPH Program Technology Support Link

In the "Getting Started" section of Canvas, there is a *Technical Support Form* where students can report any problems they are experiencing with navigation, materials, and links for this course. Please use this form to provide a description of the problem. Our MPH Online IT team will address it as soon as possible.

Required Materials

- **Books:**
 - Heyman, D. L. (Ed.). (2014). *Control of Communicable Diseases Manual* (20th ed.). Atlanta, GA: American Public Health Association.
ISBN-13: 978-0875530185
ISBN-10: 0875530184

- **Articles:**

- Please refer to the weekly reading table at the end of this syllabus for a complete list of articles or other readings.
- All course articles can be accessed via the [Course Readings](#).
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Description of Assessments and Alignment Grid

Learning Objective By the end of this course, students should be able to:	Assignment/Assessment This learning objective skill is measured by:
1. Recognize the spectrum of infectious diseases according to organ system, and clinical syndrome.	Weekly quizzes, article review and live session discussions, midterm and the final exam
2. Become familiar with the separate biology (ecology, anatomy, physiology, and pathogenesis) of pathogenic bacteria, fungal, viral, and parasitic pathogens.	Weekly quizzes, article review and live session discussions, midterm and the final exam
3. Review the basic tools and study designs of epidemiology with respect to applications in the prevention of infectious disease.	Weekly quizzes, article review and live session discussions, midterm and the final exam
4. Obtain a basic knowledge of human immunobiology and the role it can play in the protection and exacerbation of infectious disease.	Weekly quizzes, article review and live session discussions, midterm and the final exam
5. Become familiar with the methods of development and distribution of anti-infection vaccines.	Weekly quizzes, article review and live session discussions, midterm and the final exam
6. Recognize the reasons for the crisis in antibiotic development and the potential solutions.	Weekly quizzes, article review and live session discussions, midterm and the final exam
7. Appreciate the different means by which human beings can become exposed to pathogenic agents.	Weekly quizzes, article review and live session discussions, midterm and the final exam
8. Understand how strategies of prevention and intervention are based on the biology of the pathogenic agents.	Weekly quizzes, article review and live session discussions, midterm and the final exam

9. Become able to recognize patterns of infectious disease syndrome occurrence in populations, use the literature to generate possible explanations for those patterns, and evaluate methods of short-term and long-term intervention.	Weekly quizzes, article review and live session discussions, midterm and the final exam
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Description of Assessments

Quizzes and Exams

Weekly quizzes, the midterm, and the final exam will be timed exams based on the readings and instructional material. You will only be allowed to take each quiz or exam once.

Questions on the weekly quizzes include multiple choice, true/false.

Questions on the midterm include short-answer, matching, multiple choice, and true/false.

Questions on the final exam will include multiple choice and short-answer.

Group Article Reviews

Group article reviews involve presentations of responses to assigned case studies.

Groups will be assigned to present in a forum and assessed according to the following presentation rubric. Groups that are not presenting are required to respond as a group to the presentations in the forum and will be assessed according to the following response rubric.

Group Article Reviews Rubric (Presenting Group)

<i>Scoring</i>	<i>Relevance to the topic</i>	<i>Insight and application of course concepts</i>	<i>Use of evidence and support</i>
0-2 points	The posting does not directly address the question or problem being solved.	The posting does not offer any significant insight, analysis, or observation related to the topic. No knowledge or understanding is demonstrated regarding concepts and ideas pertaining to the discussion topic.	The posting does not support its claims with either evidence or argument.
3-4 points	The posting attempts to address key issues, questions, or problems, but indirectly or obliquely.	The posting attempts to offer some insight, analysis, or observation to the topic, but may still lack a full understanding of concepts and ideas being studied.	The posting generally supports claims and opinions with evidence or argument, but may leave some gaps.
5 points	The posting directly addresses key issues, questions, or problems related to the discussion's topic.	The posting offers original or thoughtful insight, analysis, or observation that demonstrates a strong grasp of concepts and ideas being studied.	The posting supports all claims and opinions with rational argument or evidence.

Class Discussion Rubric (Response to the Initial Post)

Scoring	<i>Relevance of the response to the presentation post</i>
0-2 points	The response post does not directly address the content and ideas in the post to which it is responding.
3-4 points	The response post attempts to address the post to which it is responding but is not well formulated and the concepts are not clear.
5 points	The response post directly addresses the post to which it is responding with insightful comments and content that furthers the discussion.

Individual Assignments

Individual written assignments should be submitted in PDF format, and will be assessed according to the rubric below.

Individual Assignments Rubric

Scoring	<i>Criteria</i>
85 points	Content of Each Assignment The content element will be judged on: • How effectively your material is organized. • How clearly it is written. • How accurate your writing is - i.e., ensuring all information is accurate and properly attributed/cited to reliable sources. • Quality of substance as opposed to simple opinion (unless specifically requested) • Ability to interpret the given information. • Ability to directly respond to given questions.

15 points	Grammar The grammar element will be judged on a few key elements: • Correct punctuation and spelling. • Subject/verb agreement. • Sentence construction. • Complete sentences/thoughts. • Ensuring nouns/pronouns agree. • Correct use of “that and which” and “I and me”.
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Group Discussion Assignments

Group discussion assignments will require you to work as a group and then share your work with the class in a discussion board format.

Group Discussion Assignments Rubric

Scoring	<i>Announcement/Product: Question One</i>	<i>Insight and application of course concepts</i>	<i>Use of evidence and support</i>
0-5 points	The posting does not directly address the question or problem being solved.	The posting does not offer any significant insight, analysis, or observation related to the topic. No knowledge or understanding is demonstrated regarding concepts and ideas pertaining to the discussion topic.	The posting does not support its claims with either evidence or argument.

6-7 points	The posting attempts to address key issues, questions, or problems, but indirectly or obliquely.	The posting attempts to offer some insight, analysis, or observation to the topic, but may still lack a full understanding of concepts and ideas being studied.	The posting generally supports claims and opinions with evidence or argument, but may leave some gaps.
8-10 points	The posting directly addresses key issues, questions, or problems related to the discussion's topic.	The posting offers original or thoughtful insight, analysis, or observation that demonstrates a strong grasp of concepts and ideas being studied.	The posting supports all claims and opinions with rational argument or evidence.

The **Public Service Announcement (PSA) Assignment** is one of the group discussions that you will complete for the course.

This assignment will require you to collaborate as a group to create a public service announcement for an assigned disease that notifies the target population about the outbreak, addresses what people should watch out for in regards to the disease, addresses what people need to do if infected, and addresses how the infection can be avoided. You and your group will then share your product with the class in a discussion board format.

Public Service Announcement (PSA) Assignment Rubric

Scoring	Announcement/Product: Question One	Announcement/Product: Question Two	Announcement/Product: Question Three	Announcement/Product: Question Four	Sources

20 points	Students create an accurate and interesting announcement that adequately notifies people about the outbreak.	Students create an accurate and interesting product that adequately addresses what people should watch out for in regards to the disease.	Students create an accurate and interesting product that adequately addresses what people need to do if infected by the disease.	Students create an accurate and interesting product that adequately addresses how the disease or infection can be avoided.	Students use information from the assigned and supplemental materials to support their public service announcement. Information in all source citations is correct and there are no errors in JAMA formatting.
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15 points	Students create an accurate product that adequately notifies people about the outbreak.	Students create an accurate product that adequately addresses what people should watch out for in regards to the disease.	Students create an accurate product that adequately addresses what people need to do if infected by the disease.	Students create an accurate product that adequately addresses how the disease or infection can be avoided.	Students use information from the assigned and supplemental materials to support their public service announcement. Information in all source citations is correct but there are minor errors in APA formatting.
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10 points	Students create an accurate product but it does not adequately notify people about the outbreak.	Students create an accurate product but it does not adequately address what people should watch out for in regards to the disease.	Students create an accurate product but it does not adequately address what people need to do if infected by the disease.	Students create an accurate product but it does not adequately address how the disease or infection can be avoided.	Students use information from the assigned and supplemental materials to support their public service announcement. Information in almost all source citations is correct but there are minor errors in APA formatting.
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5 points	The product is not accurate and/or does not notify people about the outbreak.	The product is not accurate and/or does not address what people should watch out for in regards to the disease.	The product is not accurate and/or does not address what people need to do if infected by the disease.	The product is not accurate and/or does not address how the disease or infection can be avoided.	Students do not use information from the assigned and supplemental materials to support their public service announcement. Information in source citations is often incorrect and/or there are major errors in APA formatting.
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Grading Breakdown

Grades will be recorded in the **CANVAS** gradebook.

Component	% of Grade
Participation (CDC Outbreak Investigation Exercise, Live Sessions)	15%
Weekly Quizzes	10%
Individual Assignments (HIV Screening, Influenza, Hepatitis, West Nile Virus, CDC Case Studies)	15%
Group Article Reviews	5%
Group Discussion Assignments (SARS, VPD, PSA Assignments)	15%
Midterm Exam	15%
Final Exam	25%
Total	100

Grading Scale

Course final grades will be determined using the following scale.

Grade	Range
A	93.0% or higher
A-	90.0%-92.9%
B+	87.0%-89.9%
B	83.0%-86.9%
B-	80.0%-82.9%
C+	77.0%-79.9%
C	73.0%-76.9%
C-	70.0%-72.9%
D	60.0%-69.9%
F	59.9% or lower

Course-specific Policies

Live (Synchronous) Sessions

This course will include frequent Live Sessions designed to further your understanding of course content and provide an opportunity for class discussion on weekly readings.

Attendance at all Live Sessions is strongly encouraged, will make up a small part of your participation grade, and will be in your best interest, as topics covered during the sessions may be included in exams or pertinent to project/paper assignments.

Review the Live Session Information page in the course where your instructor will post the times and links to the live sessions as well as links to the recorded sessions.

Please review the [Orientation to Using Zoom](#) for details on how to attend a Live Session.

Formatting Guidelines for All Written Work

- All documents should be in PDF format.
- Papers must be type-written or word-processed, double-spaced, with 12 point font.
- Make sure you keep a copy of all submitted papers.
- Please ensure that each paper is written in APA style. Refer to the APA manual. (6th Edition)
- Finally, typos and spelling errors are unacceptable at this level and reflect poorly on you.
- All page length requirements are for double-spaced pages, with 1-inch margins, in 12-point Times New Roman font.

Assignment Submission

All assignments should be submitted through CANVAS.

Grading Timeline

You will receive a grade for your assignments within two weeks of submission.

Late work

Items that are submitted late will receive 10% off the grade per day unless the student has already talked with the professor and pre-arranged that they will be absent.

Working in Groups

This course has weekly group work.

- Group discussion assignments, group public service announcement (PSA) assignments, and group article review assignments will require you to work as a group and then share your work with the class in a discussion board format.
- For the group article review assignments, one or more groups will be assigned to present case study responses to a class discussion board. Work with your group *via a third-party collaborative tool (e.g., Google Docs, SkyDrive, Skype, etc.) of your choice* to produce this assignment. Then, designate one member of your group to post the **final draft** to the discussion board for the class to review.
- These groups will be randomly assigned at the start of the course.
- Please be aware that you and your group members will be completing a Peer Feedback Survey at the end of this course.
- Once groups have been assigned, you can find groups by clicking on the 'Roster' link in the right bar of the main course page.

Netiquette

When posting on the discussion boards and chat rooms it is important to understand how to interact with one another online, netiquette. You can read more about the rules of netiquette at [15 Rules of Netiquette for Online Discussion Boards](#).

Zoom Etiquette

This course includes “live” (synchronous) sessions, which are designed to encourage active student participation and engagement with the course material. Students should come to class prepared to discuss course readings and other instructional materials. During live sessions, you are expected to respectfully, actively listen to your peers, and thoughtfully contribute your own ideas to our discussions. We encourage students to keep their cameras on for the duration of our synchronous class sessions, to the extent possible. Virtual backgrounds can be used to remove any distractions that may be present in your physical environment from the view of your peers.

Synchronous Section Recording Notice

Live sessions for this class will be recorded. Links to the live sessions will be posted on Canvas.

Course Evaluation

Two surveys will gather student opinions about the course: the mid-semester evaluation and the standard USC course evaluation survey at the end of the semester. Your opinion is valued and can make a difference in how this course is conducted; please give your honest and constructive recommendations.

This course also contains additional short weekly feedback opportunities in Canvas designed to give you an opportunity to reflect back on the week and provide more timely, specific feedback for future improvements.

Technical Skill Requirements

1. Must have a basic working knowledge of computers and Internet use.
2. Must have a working knowledge of Canvas terms, such as Modules, Groups, and Collaborations. View [Canvas Basics Guide](#).
3. Must have a working knowledge of using Canvas tools, such as how to submit an assignment and how to post in a discussion. View [Canvas Student Guide](#).

FAQs

- [How do I view my grades?](#)
- [How do I take a quiz?](#)
- [How do I reply to a discussion?](#)
- [How do I submit an assignment?](#)
- [How do I submit an assignment on behalf of a group?](#)
- [How do I view my Canvas groups?](#)
- How do I get more "how-to" help? View the [Canvas Student Guide](#).

TECHNICAL HELP

Canvas Help

Use the "Help" link on the left-side of the screen to find (1) Canvas Support Hotline, (2) Chat with Canvas Support, and (3) the Canvas Student Guide.

USC Information Technology Services (ITS)

USC ITS can assist you with connecting to the USC Network and using supported applications.

Website: <https://itservices.usc.edu/contact/> (Links to an external site.)

Support Hours

Telephone: 24 hours per day, 7 days per week

Email: Monday – Friday, 8:00 a.m. – 6:00 p.m. (not available on holidays)

Contact Information

E-Mail: consult@usc.edu

Phone: 213-740-5555

Course Schedule: A Weekly Breakdown

Week/ Module #	Topic	Readings	Assignments/ Assessments Due
1	Introduction and Basic Epidemiology	Course Readings Required • Global burden of 369 diseases and injuries in 204 countries and territories, 1990-2019 • Download Global burden of 369 diseases and injuries in 204 countries and territories, 1990-2019 Supplemental • van Panhuis WG, Grefenstette J, Jung SY, Chok NS, Cross A, Eng H, Lee BY, Zadorozhny V, Brown S, Cummings D, Burke DS. Contagious diseases in the United States from 1888 to the present. N Engl J Med. 2013 Nov 28;369(22):2152-8. • Cassini, A., Colzani, E., Pini, A., Mangen, M.J., Plass, D., McDonald, S.A., Maringhini, G., van Lier, A., Haagsma, J.A., Havelaar, A.H., Kramarz, P., Kretzschmar, M.E., and On Behalf of the BCoDE Consortium. (2018). Impact of infectious diseases on population health using incidence-based disability-adjusted life years (DALYs): results from the Burden of Communicable Diseases in Europe study, European Union and European Economic Area countries, 2009 to 2013. Euro Surveill (16). doi: 10.2807/1560-7917.ES.2018.23.16.17-00454.	1. Week 01 Assignment 01: Screening of HIV Using the HIV Rapid Testing 2. Week 01 Group A: Article Review Prep 3. Week 01 Discussion 01: Introductions 4. Week 01 Quiz 01: Weekly Quiz 5. Week 01 Extra Credit Question

2	Clinical Syndromes, Modes of Exposure, Basic Epidemiology: Study Designs	Course Readings Review the <i>Control of Communicable Diseases Manual</i> for information relevant to this week's material. - de Martel, C., Georges, D., Bray, F., Ferlay, J., & Clifford, G. M. (2020). Global burden of cancer attributable to infections in 2018: a worldwide incidence analysis. <i>The Lancet. Global health</i>, 8(2), e180–e190. https://doi-org.libproxy1.usc.edu/10.1016/S2214-109X(19)30488-7 - Fouchier, R. A., García-Sastre, A., & Kawaoka, Y. (2013). H5N1 virus: Transmission studies resume for avian flu. <i>Nature</i>, 493(7434), 609. https://doi-org.libproxy1.usc.edu/10.1038/nature11858 - Davis, H. E., McCorkell, L., Vogel, J. M., & Topol, E. J. (2023). Long COVID: major findings, mechanisms and recommendations. <i>Nature reviews. Microbiology</i>, 21(3), 133–146. https://doi-org.libproxy1.usc.edu/10.1038/s41579-022-00846-2 - Sack, D. A., Debes, A. K., Ateudjieu, J., Bwire, G., Ali, M., Ngwa, M. C., Mwaba, J., Chilengi, R., Orach, C. C., Boru, W., Mohamed, A. A., Ram, M., George, C. M., & Stine, O. C. (2021). Contrasting Epidemiology of Cholera in Bangladesh and Africa. <i>The Journal of infectious diseases</i>, 224(12 Suppl 2), S701–S709. https://doi-org.libproxy1.usc.edu/10.1093/infdis/jiab440 - GBD 2021 Antimicrobial Resistance Collaborators. Global burden of bacterial antimicrobial resistance 1990–2021: a systematic analysis with forecasts to 2050. <i>The Lancet</i>. 2024;404(10459):1199–1226. doi:10.1016/S0140-6736(24)01867-1	- Week 02 Group A: Article Review - Week 02 Group B: Article Review Prep - Week 02 Group C: Article Review Prep - Week 02 Group D: Article Review Prep - Week 02 Group E: Article Review Prep - Week 02 Quiz 01: Weekly Quiz - Week 02 Extra Credit Question
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3	Cast of Characters: Bacteria, Fungi, Viruses, and Parasites	Course Readings - Gelderblom, H. R. (1996). Structure and Classification of Viruses. In Medical Microbiology (4th ed., Chapter 41). Retrieved from https://www.ncbi.nlm.nih.gov/books/NBK8174/ - Lowry, F. (2009). Bacterial Classification. Structure and Function (PDF). Retrieved from https://www.mendeley.com/guides/apa-citation-guide - Garcia, L. F. (1999). Classification of Human Parasites, Vectors, and Similar Organisms (PDF) <i>Clinical Infectious Diseases</i>, 29(4). doi:10.1086/520425	- Week 03 Group B: Article Review - Week 03 Group C: Article Review - Week 03 Group D: Article Review - Week 03 Group E: Article Review - Week 03 Quiz 01: Weekly Quiz - Week 03 Extra Credit Question
4	Immunology	Course Readings Janeway, C. A., Travers, P., Walport, M., et al. (2001). Principles of innate and adaptive immunity. In Immunobiology: The immune system in health and disease. (5th ed., Chapter 1). Retrieved from https://www.ncbi.nlm.nih.gov/books/NBK27090/	- Week 04 Quiz 01: Weekly Quiz - Week 04 Extra Credit Question

5	Person to Person Exposure through Respiratory Droplets	Course Readings • Zumla A. 2013. <i>NEJM</i>. • Getahun H. 2015. <i>NEJM</i>. • Chaisson, R. E., & Martinson, N. A. (2008). Tuberculosis in Africa: Combating an HIV-driven crisis. The New England Journal of Medicine, 358 (11), 1089-1091.	1. Week 05 Discussion 01: SARS Discussion 2. Week 05 Discussion 02: Team Public Service Announcement (Group A) 3. Week 05 Discussion 03: Team Public Service Announcement (Group A): Critiques 4. Week 05 Assignment 01: Influenza Virus 5. Week 05 Quiz 01: Weekly Quiz 6. Week 05 Extra Credit Question
6	Person to Person: Fecal-Oral Transfer	Course Readings • Pitzer VE , Meiring J , Martineau FP , Watson CH , Kang G , Basnyat B , Baker S . The Invisible Burden: Diagnosing and Combatting Typhoid Fever in Asia and Africa. Clin Infect Dis. 2019 Oct 15;69(Supplement_5): S395-S401. • Crawford , S. E ., Ramani , S., Tate, J. E ., Parashar, U. D ., Svensson, L ., Hagbom, M ., Franco, M. A ., Greenberg, H. B ., O'Ryan, M., Kang, G., Desselberger, U., Estes, M. K .(2017). Rotavirus infection. Nature Reviews Disease Primers. Nov 9;3:17083. • DeVries, A. S., Harper, J., Murray, A., Lexau, C., Bahta, L., Christensen, J., ...Lynfield, R. (2011). Vaccine-derived poliomyelitis 12 years after infection in Minnesota. New England Journal of Medicine, (364) , 2316-2323.. Websites/Links	1. Week 06 Discussion 01: Team Public Service Announcement (Group B) 2. Week 06 Discussion 02: Team Public Service Announcement (Group B): Critiques 3. Week 06 Participation Activity: Case of the Conference Blues: E. Coli Outbreak 4. Week 06 Quiz 01: Weekly Quiz 5. Week 06 Extra Credit Question

		Level 1 Outbreak Investigation: Case of the Conference Blues: E. Coli Outbreak in the CDC's Solve the Outbreak Simulation	
7	Person to Person: Direct or Indirect Contact	Course Readings • Weber, D. J., Rutala, W. A., Miller, M. B., Huslage, K., & Sickbert-Bennet, E. (2010). Role of hospital surfaces in the transmission of emerging health care-associated pathogens: Norovirus, Clostridium difficile, and Acinetobacter species. <i>AJIC: American Journal of Infection Control</i>, 5(38), S25-S33. • Oboho, I. K., Tomczyk, S. M., Al-Asmari, A. M., Banjar, A. A., Al-Mugti, H., Alorain, i M. S., Alkhaldi, K. Z., Almohammadi, E. L., Alraddadi, B. M., Gerber, S. I., Swerdlow, D. L., Watson, J. T., Madani, T. A. (2015). 2014 MERS-CoV outbreak in Jeddah—A link to health care facilities. <i>New England Journal of Medicine</i>, 372(9), 846-54. • Briand, S., Bertherat, E., Cox, P., Formenty, P., Kieny, M. P., Myhre, J. K., Roth, C., Shindo, N., Dye, C. (2014). The international Ebola emergency. <i>New England Journal of Medicine</i>, 371(13), 1180-3 • After Ebola in West Africa: Unpredictable Risks, Preventable. <i>NEJM</i>. Aug.11, 2016.	1. Week 07 Assignment 01: VPDs 2. Week 07 Quiz 01: Weekly Quiz 3. Week 07 Extra Credit Question 4. Mid-Semester Student Survey
8	Person to Person: Genital Transfer (STDs)	Course Readings • Lundgren, J. D., Babiker, A.G., Gordin, F., Emery, S., Grund, B., Sharma, S., Avihingsanon, A., Cooper, D. A., Fätkenheuer, G., Llibre, J. M., Molina, J. M., Munderi, P., Schechter, M., Wood, R., Klingman, K. L., Collins, S., Lane, H. C., Phillips, A. N., & Neaton, J. D. (2015). Initiation of antiretroviral therapy in early asymptomatic HIV infection. <i>New England Journal of Medicine</i>, 373(9), 795-807. • Karim, R., Mack, W. J., Kono, N., Tien, P. C., Anastos, K., Lazar, J., ..., & Hodis, H. N. (2013). Gonadotropin and sex steroid levels in	1. Week 08 Discussion 01: Team Public Service Announcement (Group C) 2. Week 08 Discussion 02: Team Public Service Announcement (Group C): Critiques 3. Week 08 Group E: Article Review Prep 4. Week 08 Quiz 01: Weekly Quiz

		HIV-infected premenopausal women and their association with subclinical atherosclerosis in HIV-infected and –uninfected women in the women's interagency HIV study (WIHS). <i>The Journal of Clinical Endocrinology and Metabolism</i>, 98(4), E610-8. doi: 10.1210/jc.2012-3195 Bajunirwe , F ., Akakimpa , D ., Tumwebaze , F . P ., Abongomera , G ., Mugenyi , P . N ., & Kityo , C . M . Persistence of traditional and emergence of new structural drivers and factors for the HIV epidemic in rural Uganda: A qualitative study. (2019) <i>PLoS One</i>, 14(11).	5. Week 08 Extra Credit Question 6. Week 08: Midterm
9	Person to Person: Intravascular Transfer / Public Health Policy	Course Readings Required Group discussion: Szmunes, W., Stevens, C. E., Harley, E. J., Zang, E. A., Oleszko, W. R., William, D. C., Sadosky, R., & Kellner, A. (1980). Hepatitis B vaccine: Demonstration of efficacy in a controlled clinical trial in a high risk population in the United States. <i>New England Journal of Medicine</i>, 303, 833-841. doi: 10.1056/NEJM198010093031501. Supplemental - Khokar, N., Niazi, T. K., & Qureshi, M. O. (2013). Hepatocellular carcinoma after sustained viral response to interferon and ribavirin therapy in cirrhosis secondary to chronic hepatitis C. <i>Journal of the College of Physicians and Surgeons – Pakistan</i>, 23(10), 699-702. doi: 10.2013/JCPSP.699702. - Holmberg, S .D., Spradling, P. R., Moorman, A. C., Denniston, M. M. (2013). Hepatitis C in the United States. <i>New England Journal of Medicine</i>, 368(20):1859-61. doi: 10.1056/NEJMp1302973. - Bern C. (2015). Chagas' disease. <i>New England Journal of Medicine</i>, 373(5), 456-66.	1. Week 09 Assignment 01: Hepatitis 2. Week 09 Quiz 01: Weekly Quiz 3. Week 09 Extra Credit Question

10	Vector Borne Transmission	Course Readings - Dayan GH, Arredondo-García JL, Rivera DM, Cunha R, Deseda C, Reynales H, Costa MS, Morales-Ramírez JO, Carrasquilla G, Rey LC, Dietze R, Luz K, Rivas E, Miranda Montoya MC, Cortés Supelano M, Zambrano B, Langevin E, Boaz M, Tornieporth N, Saville M, Noriega F; CYD15 Study Group. Efficacy of a tetravalent dengue vaccine in children in Latin America. - Petersen LR. March 2016. <i>New England Journal of Medicine</i>. - Monath, TH. (2007). Dengue and yellow fever: Challenges for the development and use of vaccines. <i>New England Journal of Medicine</i>, 357, 2222-2225. doi: 10.1056/NEJMp0707161 - Barzon, L., Pacenti, M., Franchin, E., Pagni, S., Lavezzo, E., Squarzon, L., ..., Palu, G. (2013). Large human outbreak of West Nile virus infection in North-Eastern Italy in 2012. <i>Viruses</i>, 5(11), 2825-2839. doi: 10.3390/v5112825 - Chuansumrit, A., & Chaiyaratana, W. (2014). Hemostatic derangement in dengue hemorrhagic fever. <i>Thrombosis Research</i>, 133, 10-16 Websites/Links <i>Level 1 Outbreak, The Queens Killer: West Nile Virus Outbreak</i> in the CDC's Solve the Outbreak Simulation. After you've solved the outbreak, tweet your score to #PM527.	- Week 10 Discussion 01: Team Public Service Announcement (Group D) - Week 10 Discussion 02: Team Public Service Announcement (Group D): Critiques - Week 10 Assignment 01: West Nile Virus - Week 10 Participation Activity: The Queens Killer: West Nile Virus Outbreak - Week 10 Quiz 01: Weekly Quiz - Week 10 Extra Credit Question
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11	Transmission from Animals	Course Readings Madsen, J.M., Zimmerman, N.G., Timmons, J., & Tablante, N.L. (2013). Avian influenza seroprevalence and biosecurity risk factors in Maryland backyard poultry: A cross-sectional study (PDF). PLOS 8(2). doi: 10.1371/journal.pone.0056851 Websites/Links: <i>Level 1 Outbreak Investigation, Sugar Plantation Blues: Rabies Outbreak</i> in the CDC's Solve the Outbreak Simulation. Once you've solved the outbreak, tweet your score to #PM527.	1. Week 11 Discussion 01: Team Public Service Announcement (Group E) 2. Week 11 Discussion 01: Team Public Service Announcement (Group E): Critiques 3. Week 11 Participation Activity: Sugar Plantation Blues: Rabies Outbreak 4. Week 11 Quiz 01: Weekly Quiz 5. Week 11 Extra Credit Question
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12	Transmission from Soil, Air, and Water	Course Readings Review the <i>Control of Communicable Diseases Manual</i> for information relevant to this week's material. - Hotez, P. J., & Fenwick, A. (2009). Schistosomiasis in Africa: An emerging tragedy in our new global health decade. <i>PLoS Negl Trop Dis</i>, 3(9), e485. doi:10.1371/journal.pntd.0000485 - Rosen, H. E., Kimura, A. C., Crandall, J., Poe, A., Nash, J., Boetzer, J., Teclé, S., Mukhopadhyay, R., McAuley, K., Kasirye, O., Garza, A., Shahkarami, M., Chaturvedi, V., Kiang, D., Vidanes, J., McCoy, K., Barcellos, M., Derby, T., Jain, S., & Vugia, D. J. (2019). Foodborne botulism outbreak associated with commercial nacho cheese sauce from a gas station market. <i>Clinical Infectious Diseases</i>. - Tornheim, J. A., & Dooley, K. E. (2018). Challenges of TB and HIV co-treatment: updates and insights. <i>Current opinion in HIV and AIDS</i>, 13(6), 486–491. https://doi-org.libproxy1.usc.edu/10.1097/COH.0000000000000495 - Pimenoff, V. N., Gray, P., Louvanto, K., Söderlund-Strand, A., Dillner, J., & Lehtinen, M. (2023). Ecological diversity profiles of non-vaccine-targeted HPVs after gender-based community vaccination efforts. <i>Clinical and Translational Report</i>, 31(11), P1921-P1929.E3. https://doi.org/10.1016/j.chom.2023.10.001 - Kennedy, S. B., Macklin, G. R., Ross, G. M., Cavestany, R. L., Moukom, R. A., Jones, K. A. V., Mainou, B. A., Massaquoi, M. B. F., Kieh, M. W. S., & Mach, O. (2023). Poliovirus antibodies following two rounds of campaigns with a type 2 novel oral poliovirus vaccine in	- Week 12 Participation Activity: Breathless in the Midwest: Anthrax Outbreak - Week 12 Group A: Article Review Prep - Week 12 Group B: Article Review Prep - Week 12 Group C: Article Review Prep - Week 12 Group D: Article Review Prep - Week 12 Group E: Article Review Prep - Week 12 Quiz 01: Weekly Quiz - Week 12 Extra Credit Question
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		Liberia: a clustered, population-based seroprevalence survey. The Lancet Global Health, 11(6), E917-E923. https://doi.org/10.1016/S2214-109X(23)00116-X - Rodriguez, M. G., Syros, A., Rodriguez, A. E., & Serota, D. P. (2023). The Perils of Overly Sensitive Viral Load Testing for Persons With Human Immunodeficiency Virus. Open forum infectious diseases, 10(10), ofad494. https://doi.org/10.1093/ofid/ofad494 - Hegde, S., Lee, K. H., Styczynski, A., Jones, F. K., Gomes, I., Das, P., & Gurley, E. S. (2023). Potential for person-to-person transmission of henipaviruses: A systematic review of the literature. The Journal of Infectious Diseases, jiad467. Advance online publication. https://doi.org/10.1093/infdis/jiad467 - Eby, P., Peel, A. J., Hoegh, A., Madden, W., Giles, J. R., Hudson, P. J., & Plowright, R. K. (2023). Pathogen spillover driven by rapid changes in bat ecology. Nature, 613, 340–344. https://doi.org/10.1038/s41586-022-05176-0 - Keita, A. K., Koundouno, F. R., Faye, M., Düx, A., Hinzmann, J., Diallo, H., Ayoub, A., Le Marcis, F., Soropogui, B., Ifono, K., Diagne, M. M., Sow, M. S., Bore, J. A., Calvignac-Spencer, S., Vidal, N., Camara, J., Keita, M. B., Renevey, A., Diallo, A., Soumah, A. K., Millimono, S. L., Mari-Saez, A., Diop, M., Doré, A., Soumah, F. Y., Kourouma, K., Vielle, N. J., Loucoubar, C., Camara, I., Kourouma, K., Annibaldis, G., Bah, A., Thielebein, A., Pahlmann, M., Pullan, S. T., Carroll, M. W., Quick, J., Formenty, P., Legand, A., Pietro, K., Wiley, M. R., Tordo, N., Peyrefitte, C., McCrone, J. T., Rambaut, A., Sidibé, Y., Barry, M. D., Kourouma, M., Saouromou, C. D., Condé, M., Baldé, M., Povogui, M., Keita, S., Diakite, M., Bah, M. S., Sidibe, A., Diakite, D.,	
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		Sako, F. B., Traore, F. A., Ki-Zerbo, G. A., Lemey, P., Günther, S., Kafetzopoulou, L. E., Sall, A. A., Delaporte, E., Duraffour, S., Faye, O., Leendertz, F. H., Peeters, M., Toure, A., & Magassouba, N. F. (2021). Resurgence of Ebola virus in 2021 in Guinea suggests a new paradigm for outbreaks. <i>Nature</i>, 597, 539–543. https://doi.org/10.1038/s41586-021-03816-5 Websites/Links • <i>Level 1 Outbreak: Breathless in the Midwest: Anthrax Outbreak</i>: Week 12 Participation Activity: Breathless in the Midwest: Anthrax Outbreak • Complete Week 12 Assignment 01: Botulism CDC Case Study	
13	Vaccinology and Antibiotics	Course Readings • Amanna, I. J., Carlson, N. E., & Slifka, M. K. (2007). Duration of humoral immunity to common viral and vaccine antigens. <i>New England Journal of Medicine</i>, 357, 1903-15. doi: 10.1056/NEJMoa066092. • Spelberg, B., Guidos, R., Gilbert, D., Bradley, J., Boucher, H. W., Scheld W.M., ..., Edwards, J. (2008). The epidemic of antibiotic-resistant infections: a call to action for the medical community from the Infectious Diseases Society of America. <i>Clinical Infectious Diseases</i>, 46(2), 155-164. doi: 10.1086/524891. • Schwartz, J. L. (2014, November). <i>NEJM</i>.	1. Week 13 Group A: Article Review 2. Week 13 Group B: Article Review 3. Week 13 Group C: Article Review 4. Week 13 Group D: Article Review 5. Week 13 Group E: Article Review 6. Week 13 Quiz 01: Weekly Quiz 7. Week 13 Extra Credit Question

14	Outbreak Investigations, Disease Control	Course Readings Modlin, J. (2014, October 16). Achieving and maintaining polio eradication—New strategies. <i>The New England Journal of Medicine</i>, 371(16), 1476-1479. Websites/Links Complete the Week 14 Assignment 01: Salmonella Outbreak Case Study	1. Week 14 Assignment 01: Salmonella Outbreak Case Study
15	Review	No readings this week. Please complete Course Evaluations.	1. Week 15 Final Exam 2. Peer Evaluation

Statement on Academic Conduct and Support Systems

Academic Conduct:

Plagiarism – presenting someone else’s ideas as your own, either verbatim or recast in your own words – is a serious academic offense with serious consequences. Please familiarize yourself with the discussion of plagiarism in SCampus in Part B, Section 11, “Behavior Violating University Standards” policy.usc.edu/scampus-part-b. Other forms of academic dishonesty are equally unacceptable. See additional information in SCampus and university policies on [Research and Scholarship Misconduct](#).

Students and Disability Accommodations:

USC welcomes students with disabilities into all of the University’s educational programs. The Office of Student Accessibility Services (OSAS) is responsible for the determination of appropriate accommodations for students who encounter disability-related barriers. Once a student has completed the OSAS process (registration, initial appointment, and submitted documentation) and accommodations are determined to be reasonable and appropriate, a Letter of Accommodation (LOA) will be available to generate for each course. The LOA must be given to each course instructor by the student and followed up with a discussion. This should be done as early in the semester as possible as accommodations are not retroactive. More information can be found at osas.usc.edu. You may contact OSAS at (213) 740-0776 or via email at osasfrontdesk@usc.edu.

Support Systems:

Counseling and Mental Health - (213) 740-9355 – 24/7 on call

studenthealth.usc.edu/counseling

Free and confidential mental health treatment for students, including short-term psychotherapy, group counseling, stress fitness workshops, and crisis intervention.

National Suicide Prevention Lifeline - 1 (800) 273-8255 – 24/7 on call

suicidepreventionlifeline.org

Free and confidential emotional support to people in suicidal crisis or emotional distress 24 hours a day, 7 days a week.

Relationship and Sexual Violence Prevention Services (RSVP) - (213)

740-9355(WELL), press “0” after hours – 24/7 on call

studenthealth.usc.edu/sexual-assault

Free and confidential therapy services, workshops, and training for situations related to gender-based harm.

Office for Equity, Equal Opportunity, and Title IX (EEO-TIX) - (213) 740-5086

eeotix.usc.edu

Information about how to get help or help someone affected by harassment or discrimination, rights of protected classes, reporting options, and additional resources for students, faculty, staff, visitors, and applicants.

Reporting Incidents of Bias or Harassment - (213) 740-5086 or (213) 821-8298

usc-advocate.symplcity.com/care_report

Avenue to report incidents of bias, hate crimes, and microaggressions to the Office for Equity, Equal Opportunity, and Title for appropriate investigation, supportive measures, and response.

The Office of Student Accessibility Services (OSAS) - (213) 740-0776

osas.usc.edu

OSAS ensures equal access for students with disabilities through providing academic accommodations and auxiliary aids in accordance with federal laws and university policy.

USC Campus Support and Intervention - (213) 821-4710

campussupport.usc.edu

Assists students and families in resolving complex personal, financial, and academic issues adversely affecting their success as a student.

Diversity, Equity and Inclusion - (213) 740-2101

diversity.usc.edu

Information on events, programs and training, the Provost's Diversity and Inclusion Council, Diversity Liaisons for each academic school, chronology, participation, and various resources for students.

USC Emergency - UPC: (213) 740-4321, HSC: (323) 442-1000 – 24/7 on call

dps.usc.edu, emergency.usc.edu

Emergency assistance and avenue to report a crime. Latest updates regarding safety, including ways in which instruction will be continued if an officially declared emergency makes travel to campus infeasible.

USC Department of Public Safety - UPC: (213) 740-6000, HSC: (323) 442-120 – 24/7 on call

dps.usc.edu

Non-emergency assistance or information.

Office of the Ombuds - (213) 821-9556 (UPC) / (323-442-0382 (HSC)

ombuds.usc.edu

A safe and confidential place to share your USC-related issues with a University Ombuds who will work with you to explore options or paths to manage your concern.

Occupational Therapy Faculty Practice - (323) 442-3340 or otfp@med.usc.edu

chan.usc.edu/otfp

Confidential Lifestyle Redesign services for USC students to support health promoting habits and routines that enhance quality of life and academic performance.